Autoencoders & GANs Report

Building Variational Autoencoders for anomaly detection and Conditional DCGANs for synthetic data generation. Two semiconductor inspection datasets with MLflow experiment tracking.

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Project 1 — Sensor Anomaly VAE

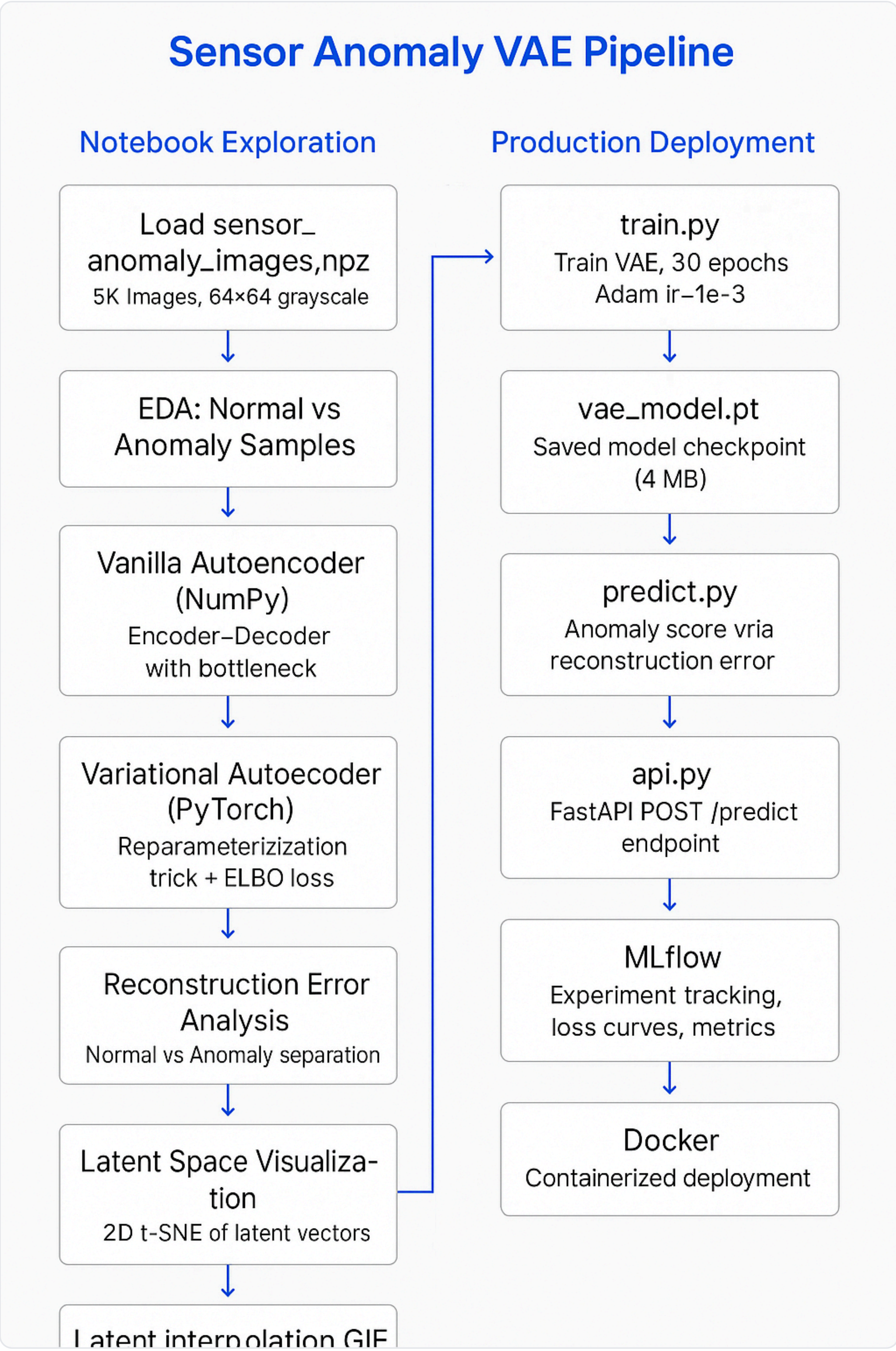
Overview

A Variational Autoencoder trained on 5,000 industrial sensor images (4,000 normal + 1,000 anomalous) to detect manufacturing defects via reconstruction error. The VAE learns to compress normal patterns into a 32-dimensional latent space; anomalous inputs produce significantly higher reconstruction error, enabling unsupervised anomaly detection without labeled fault data.

Dataset

Property	Value
File	sensor_anomaly_images.npz
Samples	5,000 (4,000 normal + 1,000 anomaly)
Resolution	64 × 64 grayscale
Pixel range	0.11 – 1.0 (float32)
Task	Unsupervised anomaly detection

Flowchart



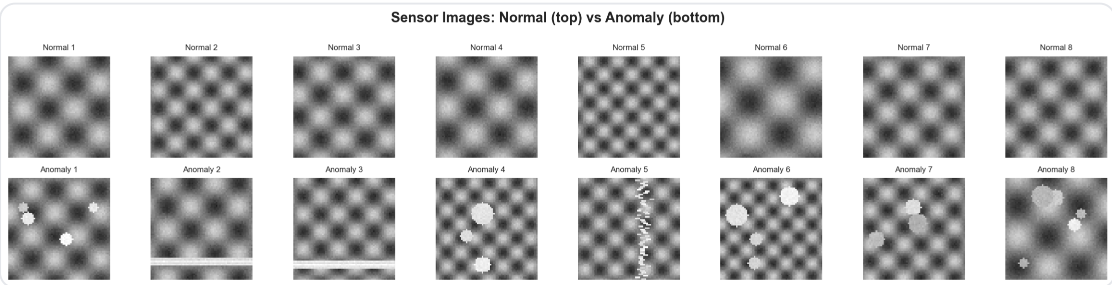
Sensor Anomaly VAE Pipeline

Results

6.66x	0.0010	0.0067
Anomaly Separation Ratio	Normal Recon Error	Anomaly Recon Error

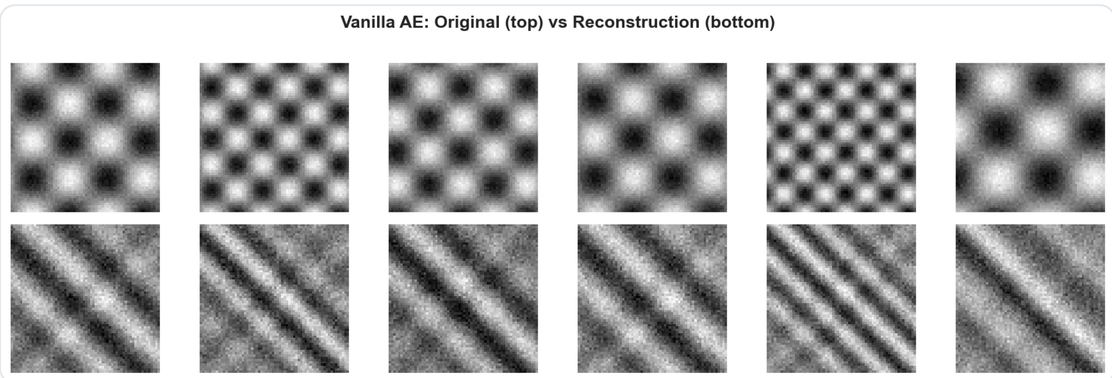
Metric	Value
Architecture	Conv2d 1→32→64→128, Latent 32-dim
Training	30 epochs, Adam lr=1e-3, ELBO loss
Normal reconstruction error	0.001006
Anomaly reconstruction error	0.006694
Separation ratio	**6.66x**
Model size	4.0 MB (vae_model.pt)

Sensor Samples



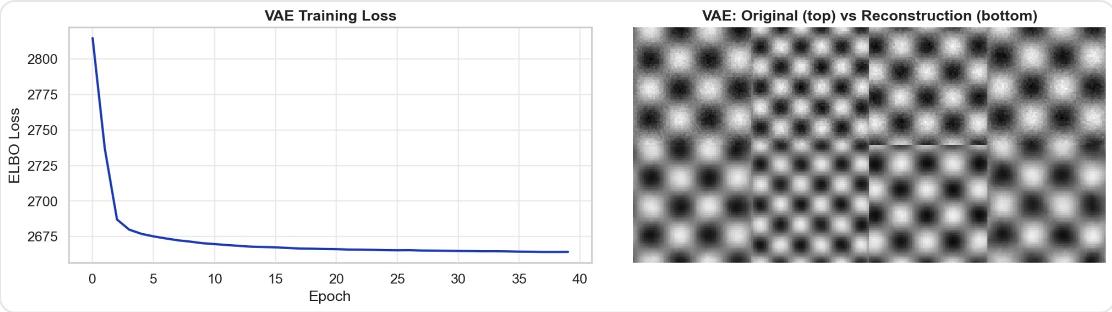
Normal vs Anomaly sensor image samples

Vanilla Autoencoder Reconstruction



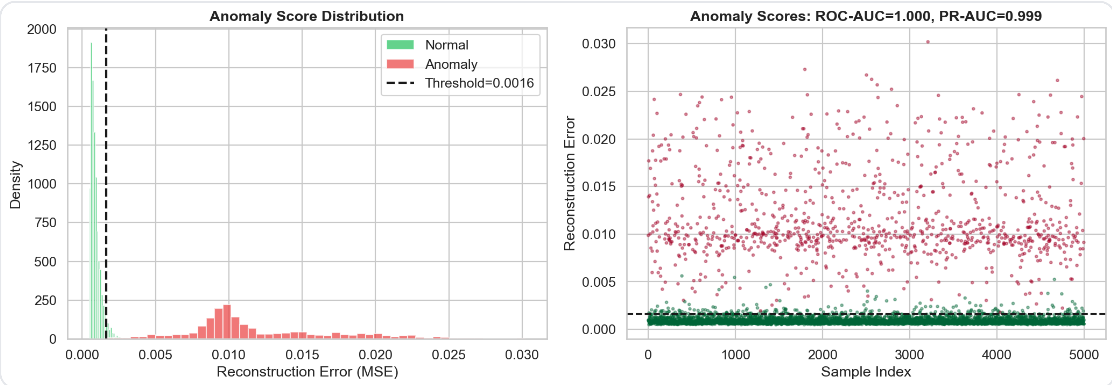
Vanilla AE: input vs reconstruction

VAE Training



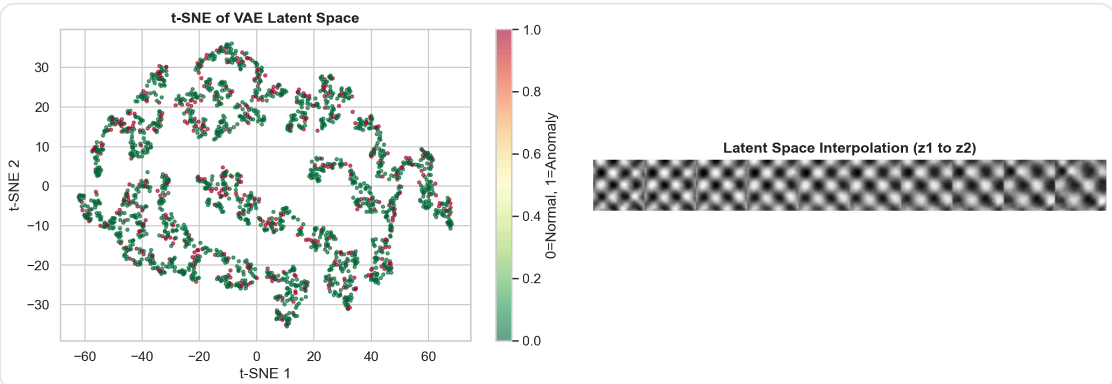
VAE training loss convergence

Anomaly Detection



Reconstruction error: normal vs anomaly separation

Latent Space



2D visualization of VAE latent space

Project 2 — Wafer Defect DCGAN

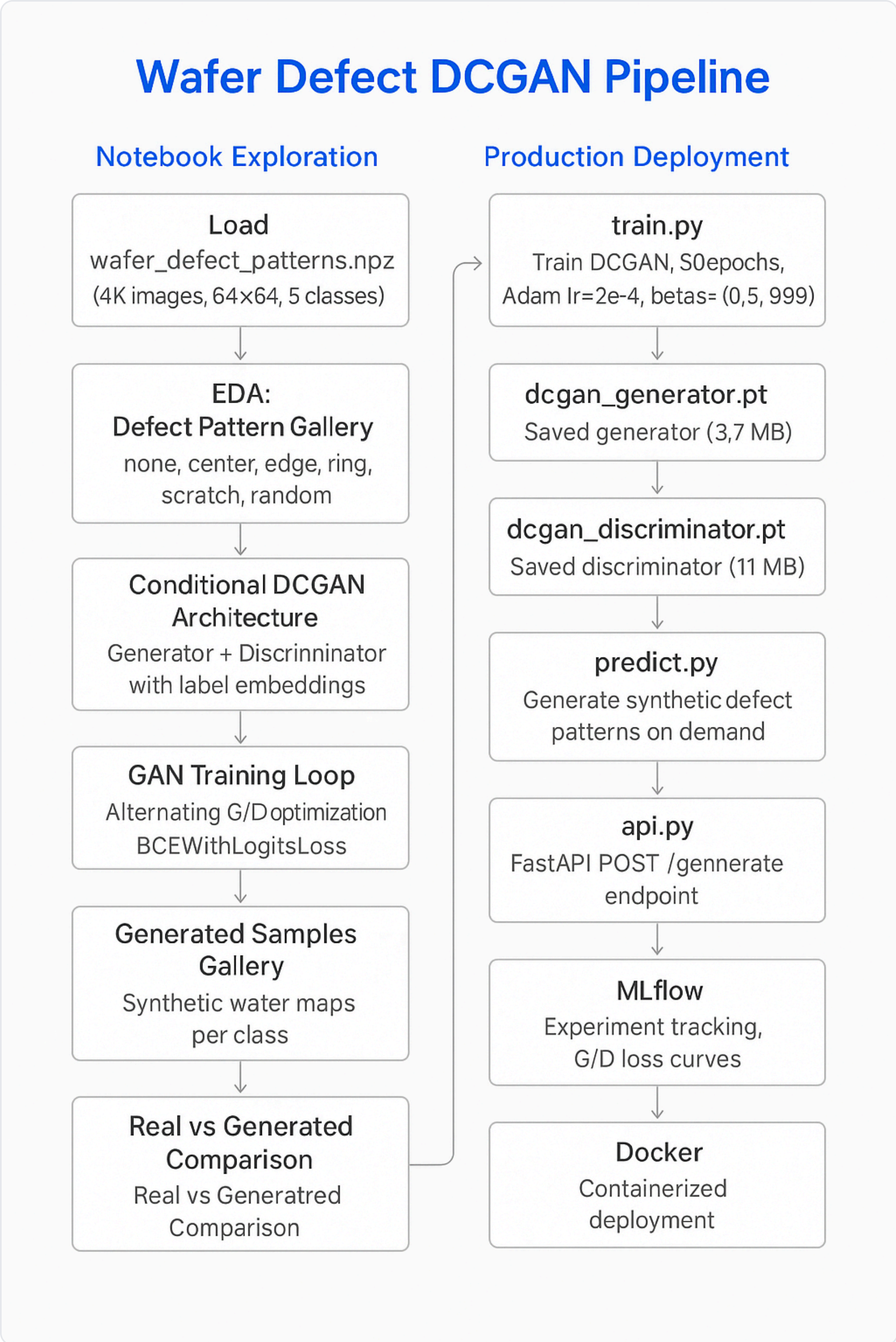
Overview

A Conditional Deep Convolutional GAN trained on 4,000 wafer map images across 5 defect pattern classes. The generator learns to produce realistic synthetic wafer maps conditioned on class labels, addressing the chronic class imbalance problem in semiconductor yield engineering. Label embeddings enable on-demand generation of specific defect types.

Dataset

Property	Value
File	wafer_defect_patterns.npz
Samples	4,000 (800 per class, balanced)
Resolution	64 × 64 grayscale
Classes	none, center, edge_ring, scratch, random
Task	Conditional image generation

Flowchart

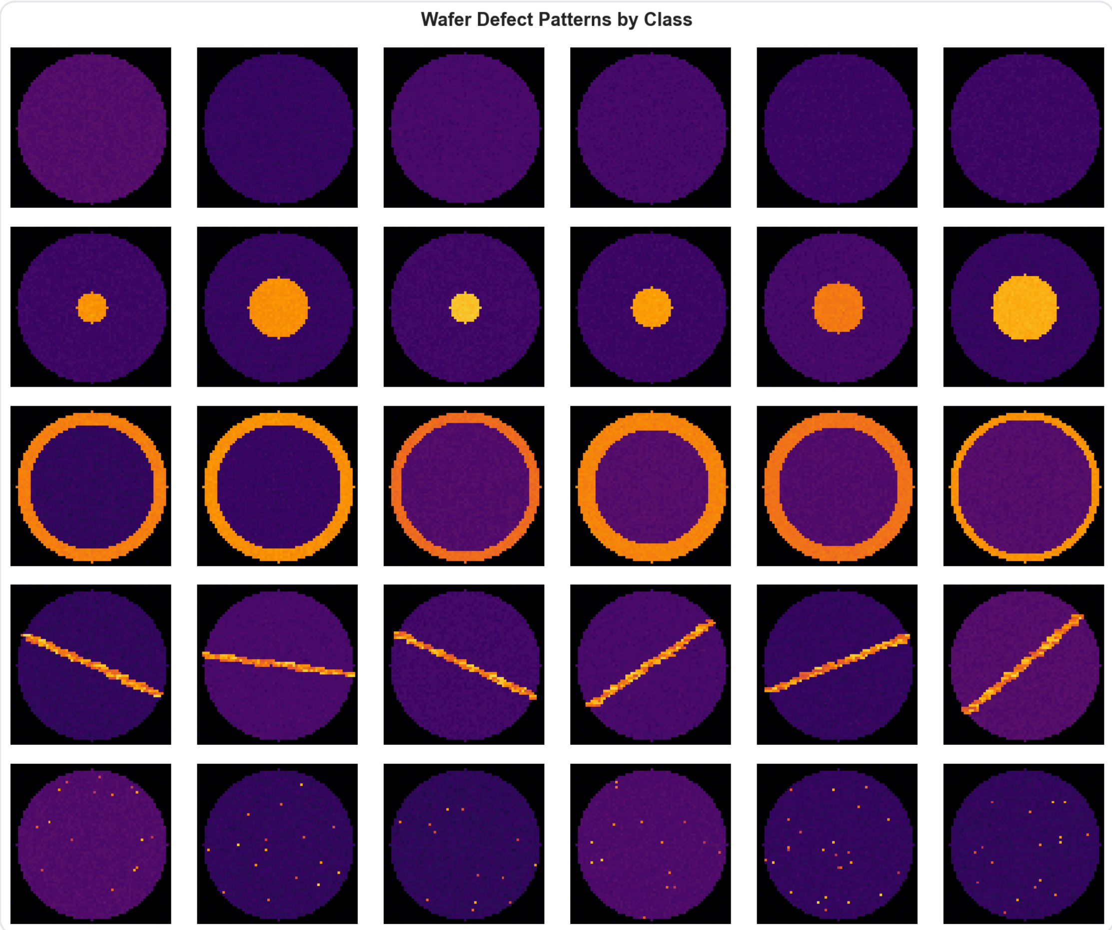


Wafer Defect DCGAN Pipeline

Results

Metric	Value
Generator	ConvTranspose2d 256→128→64→32→1, Tanh output
Discriminator	Conv2d 2→64→128→256→512, Conditional (label embedding)
Training	50 epochs, Adam lr=2e-4, betas=(0.5, 0.999)
Final D_loss	0.1737
Final G_loss	2.8102
Generator size	3.7 MB (dcgan_generator.pt)
Discriminator size	11 MB (dcgan_discriminator.pt)

Wafer Defect Samples



Wafer defect pattern samples across 5 classes

Appendix

Tech Stack

Tool	Version	Purpose
Python	3.12	Core language
PyTorch	2.0+	VAE, DCGAN training and inference
NumPy	1.24+	Data loading, array operations
scikit-learn	1.3+	Metrics
Matplotlib / seaborn	3.7+ / 0.12+	Visualization
MLflow	2.5+	Experiment tracking
FastAPI	0.100+	REST API serving
Docker	24+	Containerized deployment

Key Learnings

Concept	Key Insight
Variational Autoencoder	Reparameterization trick enables backprop through stochastic sampling; ELBO = Reconstruction + KL divergence
Anomaly Detection via VAE	6.66x separation between normal and anomaly reconstruction error enables robust unsupervised fault detection
Conditional DCGAN	Label embeddings give fine-grained control over generated content; essential for targeted data augmentation
GAN Training Dynamics	Balanced D/G loss is key; BCEWithLogitsLoss + Adam with low beta1 stabilizes training
Latent Interpolation	Smooth transitions in VAE latent space confirm the model has learned meaningful continuous representations
MLflow Tracking	Per-epoch loss logging enables experiment comparison; artifact logging preserves model provenance